

1. AI Resume Screening System

Project Summary

Build a web application that automatically evaluates resumes against a job description. The system extracts text from uploaded resumes, identifies relevant skills, and calculates a match score.

Key Features

- Upload PDF/DOCX resumes
- Extract resume content
- Compare with job description
- Calculate match percentage
- Highlight missing skills
- Rank multiple candidates

AI Concepts

- Natural Language Processing (NLP)
- Text Similarity
- Skill Extraction
- TF-IDF / Embeddings

Tech Stack

Frontend

- HTML/CSS
- Bootstrap or Streamlit

Backend

- Python
- Flask or FastAPI

AI/ML Libraries

- Scikit-learn
- NLTK / SpaCy
- Sentence Transformers (optional)

Database

- SQLite

Other Libraries

- PyPDF2
- python-docx
- Pandas

Expected Output

Resume Match Score: 85%

Matched Skills:

- ✓ Python
- ✓ SQL
- ✓ Machine Learning

Missing Skills:

- ✗ Docker
 - ✗ AWS
-

2. YouTube Video Summarizer

Project Summary

Create an application that accepts a YouTube URL, extracts the video's transcript, and generates a concise summary using AI.

Key Features

- Enter YouTube URL
- Extract transcript automatically
- Generate summary
- Show key points
- Export summary as PDF

AI Concepts

- Text Summarization
- Large Language Models (LLMs)
- Prompt Engineering
- NLP

Tech Stack

Frontend

- Streamlit

Backend

- Python

AI Services

- OpenAI API / Gemini API

Libraries

- youtube-transcript-api
- LangChain (optional)
- Transformers (optional)
- Pandas

Database (Optional)

- SQLite

Project Flow

YouTube URL

↓

Transcript Extraction

↓

AI Summarization

↓

Summary + Key Takeaways

Sample Output

Video Summary

- Introduction to Python
- Variables and Data Types
- Loops and Functions
- Best Practices

Duration: 15 mins

Summary Length: 250 words

3. AI Email Spam Detector

Project Summary

Develop a machine learning model that classifies emails as Spam or Not Spam based on their content.

Key Features

- Enter email content
- Detect spam instantly
- Display confidence score
- Train model on real email dataset
- Show important keywords influencing prediction

AI Concepts

- Text Classification
- Supervised Learning
- Feature Extraction
- Machine Learning Pipeline

Tech Stack

Frontend

- Streamlit

Backend

- Python

Machine Learning

- Scikit-learn

Libraries

- Pandas
- NumPy
- NLTK
- Joblib

Algorithms

- Naive Bayes
- Logistic Regression

Dataset

- SMS Spam Collection Dataset
- Enron Email Dataset

Project Flow

Email Text
↓
Text Cleaning
↓
TF-IDF Vectorization
↓
ML Model
↓
Spam / Not Spam

Sample Output

Email Content:
"Congratulations! You won \$10,000."

Prediction:
SPAM

Confidence:
98.2%